

1. Which of the following cell organelles is responsible for cellular respiration and ATP generation?

- A) Lysosome
- B) Mitochondrion
- C) Ribosome
- D) Endoplasmic Reticulum

2. What is the primary component of the plant cell wall?

- A) Peptidoglycan
- B) Chitin
- C) Cellulose
- D) Glycogen

3. Protein synthesis in a eukaryotic cell takes place in:

- A) The nucleus
- B) The Golgi apparatus
- C) Ribosomes
- D) The vacuole [\[1, 2\]](#)

4. Which organelle is often referred to as the "suicide bag" of the cell due to its hydrolytic enzymes?

- A) Peroxisome
- B) Lysosome
- C) Centrosome
- D) Chloroplast

5. Which of the following structures is present in animal cells but strictly absent in plant cells?

- A) Mitochondria
 - B) Cell membrane
 - C) Centrosome
 - D) Vacuole
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Section B: Biotechnology

6. Which of the following is commonly used as a cloning vector in plant biotechnology?

- A) pBR322
- B) Ti plasmid
- C) BAC (Bacterial Artificial Chromosome)
- D) YAC (Yeast Artificial Chromosome) [\[1\]](#)

7. Restriction enzymes were originally discovered and isolated from:

- A) Fungi
- B) Viruses
- C) Bacteria
- D) Protozoa [\[1\]](#)

8. Which technique is used to transfer and separate DNA fragments from an electrophoretic gel onto a nitrocellulose membrane?

- A) Southern blotting
- B) Northern blotting

- C) Western blotting
- D) PCR

9. What is the primary function of DNA ligase in genetic engineering?

- A) To cut DNA at specific sites
- B) To join DNA fragments together
- C) To synthesize RNA primers
- D) To amplify DNA sequences [[1](#), [2](#)]

10. Which of the following enzymes is used in the Polymerase Chain Reaction (PCR) to synthesize new DNA strands?

- A) RNA polymerase
- B) Taq polymerase
- C) Restriction endonuclease
- D) Reverse transcriptase